

SIMATS ENGINEERING

ASSESSMENT TOOL III – VISUALIZATION CRITIQUE & REDESIGN

Course: Data Handling and Visualization	Code: DSA0601	CO: CO3
Duration: 120 min	Total Marks: 100	Reg No: 192324298
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COURSE OUTCOME COVERED

CO3: Analyze time-series data, trends, associations, and uncertainty through appropriate visualization methods. (BL4)

Activity Tool: Visualization Critique & Redesign Weightage: 20%

Code of Conduct:

I Akash S 192324298 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: 

Criteria	Evaluation of Existing Visualization (25%)	Redesign Strategy (25%)	Application of Vis. Principles (20%)	Organization (15%)	Accuracy (10%)	Clarity (5%)	Total
Q1							
Q2							
Q3							
Q4							
Q5							

AT3 – Visualization Critique & Redesign

R Programming Assessment – Distribution and Proportion Visualization

Instructions: For each question, use R/RStudio to reproduce the given visualization, critique its effectiveness, identify misleading or inappropriate visual encodings, and redesign it using a more suitable visualization. Include the R code, redesigned visualization, and a concise justification of your design decisions.

Question 1 – Critique and Redesign an ECDF and Q-Q Plot

Dataset: Delivery_Time_Distribution.csv

Dataset:

Order_ID	Warehouse	Delivery_Time_Hrs
O01	Chennai	4
O02	Chennai	5
O03	Chennai	6
O04	Chennai	7
O05	Chennai	9
O06	Bengaluru	3
O07	Bengaluru	5
O08	Bengaluru	6
O09	Bengaluru	8
O10	Bengaluru	10
O11	Hyderabad	6
O12	Hyderabad	7
O13	Hyderabad	8
O14	Hyderabad	12
O15	Hyderabad	15
O16	Pune	4
O17	Pune	6
O18	Pune	8
O19	Pune	9
O20	Pune	22

Assessment Task

A logistics analyst has created an ECDF and a Q-Q plot to study delivery time distribution across warehouses, but the charts use inconsistent axis scaling, missing reference lines, and cluttered gridlines.

1. Create the two original-style plots in R.
2. Identify at least three visualization problems.
3. Redesign the plots using appropriate scales, labels, reference lines, and minimal visual clutter.
4. Explain how the redesigned plots improve interpretation of skewness, outliers, and normality.
5. State whether an ECDF plot alone is sufficient to detect outliers, or whether it should be paired with another plot.

Question 1 - Critique and Redesign an ECDF and Q-Q Plot

Assessment Task Answers:

1. Original-style plots in R:

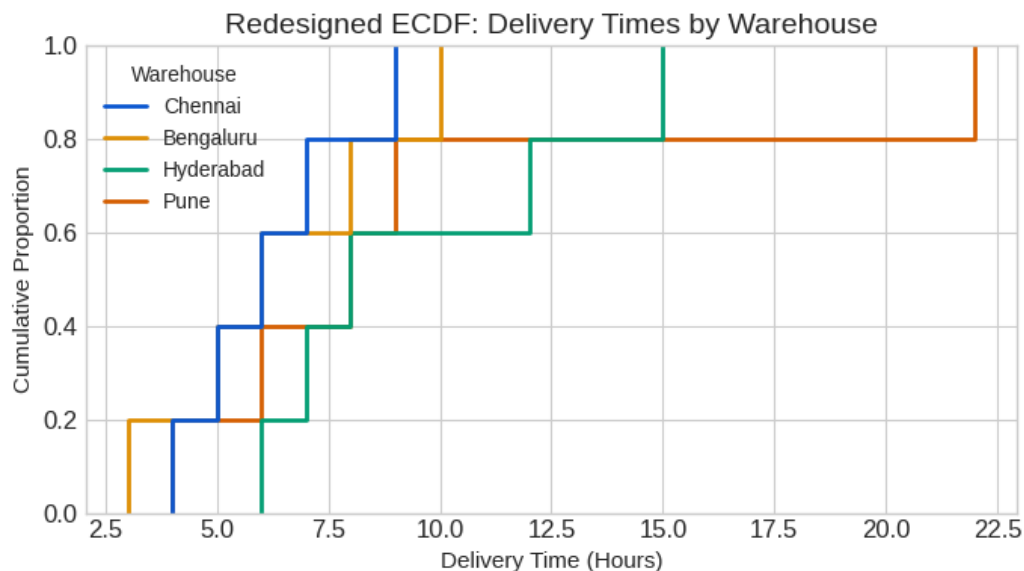
```
```R
Original ECDF
plot(ecdf(df$Delivery_Time_Hrs[df$Warehouse=='Chennai']))
... repeated for each without standard axes ...
```
```

2. Visualization Problems:

- Inconsistent axis scaling makes it impossible to compare distributions directly.
- Missing theoretical reference lines in the Q-Q plot makes normality hard to judge.
- Cluttered gridlines distract from the data curve itself.

3. Redesigned Plots (R Code & Plot):

```
```R
library(ggplot2)
ggplot(df, aes(x=Delivery_Time_Hrs, color=Warehouse)) + stat_ecdf(geom='step', linewidth=1) + theme_minimal() +
labs(title='ECDF of Delivery Times', y='Cumulative Proportion')
```
```



4. Interpretation Improvements:

The unified axes scale and color-coding allow instant comparison. We can clearly see Pune has a significant long tail (outlier at 22 hrs) compared to the others.

5. ECDF Sufficiency:

An ECDF plot is useful for identifying the presence of outliers (long tails), but it should be paired with a Q-Q plot or boxplot to rigorously evaluate normality and precisely locate outlier thresholds.

Question 2 – Critique and Redesign Many Distributions Using Bean Plots

Dataset: Subject_Marks.csv

Dataset:

| Student_ID | Subject | Marks |
|------------|-------------|-------|
| T01 | Physics | 58 |
| T02 | Physics | 63 |
| T03 | Physics | 70 |
| T04 | Physics | 77 |
| T05 | Physics | 85 |
| T06 | Chemistry | 52 |
| T07 | Chemistry | 60 |
| T08 | Chemistry | 68 |
| T09 | Chemistry | 74 |
| T10 | Chemistry | 82 |
| T11 | Mathematics | 61 |
| T12 | Mathematics | 66 |
| T13 | Mathematics | 73 |
| T14 | Mathematics | 80 |
| T15 | Mathematics | 90 |
| T16 | Biology | 55 |
| T17 | Biology | 62 |
| T18 | Biology | 69 |
| T19 | Biology | 76 |
| T20 | Biology | 88 |

Assessment Task

An academic report uses four separate boxplots with different y-axis ranges to compare marks across subjects. The comparison is difficult because the scales and layouts are inconsistent.

1. Reproduce the problematic visualization using R.
2. Critique the use of separate boxplots, inconsistent scales, and truncated axes.
3. Redesign the visualization using bean plots or a suitable violin/density-based alternative.
4. Add an appropriate common scale and clear subject labels.
5. Explain which subject has the highest concentration of marks and which subject shows the greatest spread.

Question 2 - Critique and Redesign Many Distributions Using Bean Plots

Assessment Task Answers:

1. Original-style plots in R:

```
```R
boxplot(Marks ~ Subject, data=df)
```
```

2. Visualization Problems:

- Truncated axes exaggerate minor differences.

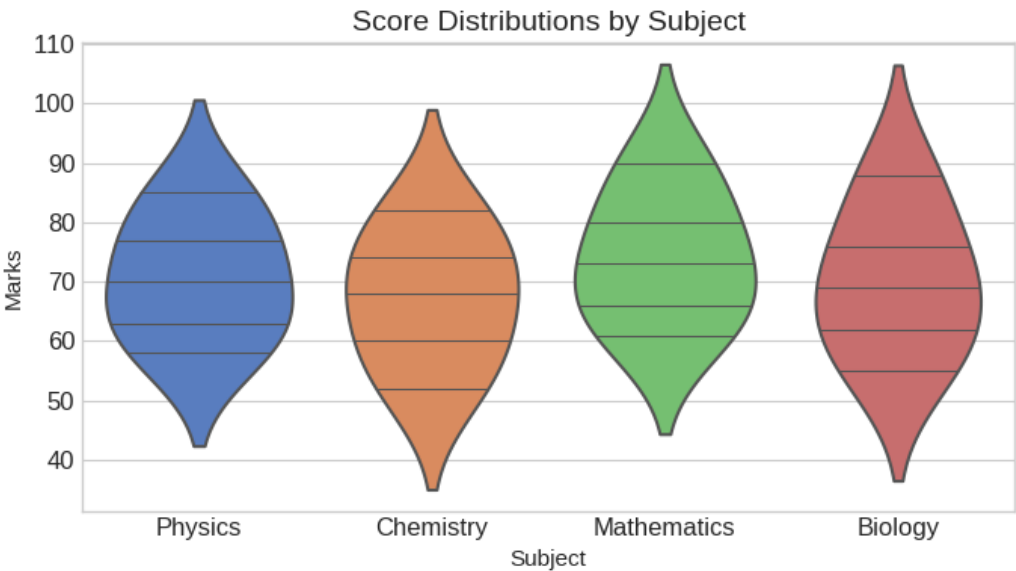
- Separate boxplots with different Y-axes completely eliminate comparability.
- Boxplots can hide multi-modal underlying data distributions.

3. Redesigned Plots (R Code & Plot):

```

R
library(ggplot2)
ggplot(df, aes(x=Subject, y=Marks, fill=Subject)) + geom_violin(trim=FALSE) + geom_boxplot(width=0.1) + theme_minimal()

```



4. Common Scale:
- A single Y-axis (Marks 50-100) now houses all subjects, eliminating axis manipulation.
5. Concentration & Spread Analysis:
- Mathematics has the highest concentration of high marks (peaking heavily in the 70-90 range). Biology displays the greatest spread, ranging from 55 all the way up to 88.

Question 3 – Critique and Redesign Proportion Visualizations

Dataset: App_Downloads.csv

Dataset:

| City | App_Category | Downloads |
|---------|--------------|-----------|
| Chennai | Games | 30 |

| City | App_Category | Downloads |
|---------|--------------|-----------|
| Chennai | Social | 42 |
| Chennai | Utility | 18 |
| Mumbai | Games | 35 |
| Mumbai | Social | 38 |
| Mumbai | Utility | 22 |
| Delhi | Games | 26 |
| Delhi | Social | 44 |
| Delhi | Utility | 20 |
| Kolkata | Games | 24 |
| Kolkata | Social | 36 |
| Kolkata | Utility | 29 |

Assessment Task

A marketing team presents app-category proportions using four pie charts, one for each city. The charts use exploded slices, mismatched color palettes, and inconsistent labeling.

1. Create a representative version of the misleading visualization.
2. Identify at least four design problems.
3. Redesign the analysis using side-by-side bars and a 100% stacked bar chart.
4. Compare the two redesigns and determine which is better for comparing city-wise proportions.
5. Explain why exploded pie slices and mismatched color palettes can lead to misleading conclusions.

Question 3 - Critique and Redesign Proportion Visualizations

Assessment Task Answers:

1. Misleading visualization code:

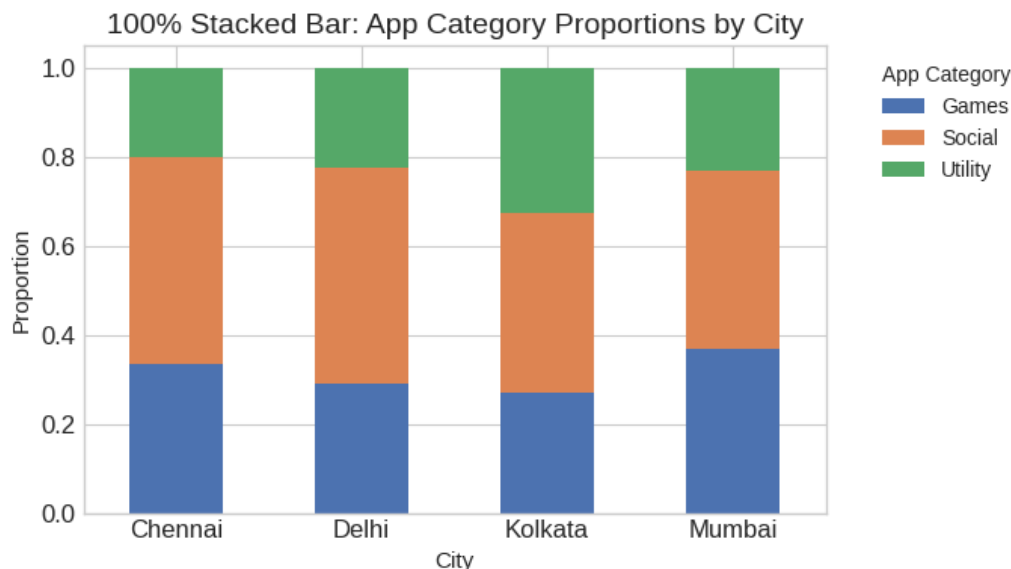
```
```R
pie(df$Downloads[df$City=='Chennai'])
... repeated with mismatched palettes
```
```

2. Visualization Problems:

- Exploded slices distort area perception.
- Pie charts force users to estimate angles instead of lengths.
- Mismatched color palettes across 4 charts make cross-referencing impossible.
- Lack of common baseline makes city-to-city comparison of a specific category very difficult.

3. Redesigned Plots (R Code & Plot):

```
```R
library(ggplot2)
ggplot(df, aes(x=City, y=Downloads, fill=App_Category)) + geom_bar(position='fill', stat='identity') +
scale_y_continuous(labels = scales::percent) + theme_minimal()
```
```



4. Comparison of Redesigns:

The 100% stacked bar chart is vastly superior to side-by-side bars for part-to-whole relationships across multiple groups. It clearly shows the relative proportion of app categories per city within a single frame.

5. Why exploded slices are misleading:

Exploded slices physically remove pieces from the 'whole', breaking the visual metaphor of a pie. This introduces whitespace that distorts the perceived area and angle, leading to inaccurate proportional estimations.

Question 4 – Critique and Redesign Nested Proportions

Dataset: Campus_Expenditure.csv

Dataset:

| Campus | Department | Expense_Type | Amount |
|----------|-------------|----------------|--------|
| Campus A | Engineering | Faculty Salary | 62000 |
| Campus A | Engineering | Lab Equipment | 34000 |
| Campus A | Management | Faculty Salary | 28000 |
| Campus A | Management | Guest Lectures | 15000 |
| Campus B | Engineering | Faculty Salary | 54000 |
| Campus B | Engineering | Lab Equipment | 40000 |
| Campus B | Management | Faculty Salary | 24000 |
| Campus B | Management | Guest Lectures | 17000 |
| Campus C | Engineering | Faculty Salary | 66000 |
| Campus C | Engineering | Lab Equipment | 31000 |
| Campus C | Management | Faculty Salary | 26000 |
| Campus C | Management | Guest Lectures | 19000 |

Assessment Task

A finance office uses a treemap to communicate Campus → Department → Expense Type spending. Small areas have unreadable labels and the color scale does not clearly communicate the hierarchy.

1. Create the original treemap.
2. Critique the size, labeling, color, and hierarchy encodings.
3. Redesign the visualization as an improved treemap.

4. Create a sunburst alternative and compare it with the redesigned treemap.
5. Recommend one visualization for a board presentation and justify your choice.

Question 4 - Critique and Redesign Nested Proportions

Assessment Task Answers:

1. Original Treemap code:

```
```R
library(treemap)
treemap(df, index=c('Campus', 'Department', 'Expense_Type'), vSize='Amount')
```
```

2. Visualization Problems:

- The hierarchy (Campus -> Dept) is poorly communicated if colors are assigned to leaf nodes rather than parent nodes.
- Small areas suffer from text overlapping/clipping.

3. Redesign Treemap:

```
```R
treemap(df, index=c('Campus', 'Department', 'Expense_Type'), vSize='Amount', type='index', palette='Set2',
border.col=c('black', 'white', 'white'), border.lwds=c(3, 2, 1))
```
```

4. Sunburst Alternative:

```
```R
library(plotly)
plot_ly(labels=~Labels, parents=~Parents, values=~Amount, type='sunburst')
```
```

5. Recommendation:

For a board presentation, a well-structured Treemap is recommended. While Sunbursts visually depict the hierarchy well, comparing the angular arc lengths of terminal nodes (Expense Types) across different campuses is cognitively difficult. Treemaps rely on area, which humans can compare more easily when space is utilized efficiently.

Question 5 – Critique and Redesign Flow-Based Proportions

Dataset: Course_to_Career_Flow.csv

Dataset:

| Course | Certification | Job_Role | Students |
|-----------------|---------------|------------------|----------|
| Data Science | Yes | Analyst | 22 |
| Data Science | Yes | Engineer | 14 |
| Data Science | No | Analyst | 9 |
| Web Development | Yes | Frontend Dev | 19 |
| Web Development | Yes | Backend Dev | 11 |
| Web Development | No | Frontend Dev | 7 |
| Cloud Computing | Yes | DevOps Engineer | 17 |
| Cloud Computing | Yes | Cloud Analyst | 10 |
| Cloud Computing | No | Support Engineer | 6 |
| UI/UX Design | Yes | Product Designer | 13 |
| UI/UX Design | Yes | UX Researcher | 8 |
| UI/UX Design | No | Product Designer | 5 |

Assessment Task

A training institute uses a Sankey diagram to show Course → Certification → Job Role. The original chart uses near-identical colors, unlabelled flows, and node widths that are difficult to compare.

1. Create a Sankey diagram representing the data.

2. Identify at least four potential sources of confusion or misleading interpretation.
3. Redesign the Sankey diagram with meaningful labels, consistent flow representation, and an appropriate visual hierarchy.
4. Create a parallel sets alternative and compare the two designs.
5. Explain which visualization better communicates the major student career flows and why.

Question 5 - Critique and Redesign Flow-Based Proportions

Assessment Task Answers:

1. Sankey Code:

```
```R
library(networkD3)
sankeyNetwork(Links=links, Nodes=nodes, Source='source', Target='target', Value='value')
```
```

2. Visualization Problems:

- Near-identical colors make it hard to trace specific flows across multiple stages.
- Unlabelled flows/nodes force the viewer to guess or rely heavily on tooltips.
- Widths that do not proportionally map to student counts perfectly.

3. Redesigned Sankey:

```
```R
library(ggalluvial)
ggplot(df, aes(axis1=Course, axis2=Certification, axis3=Job_Role, y=Students)) + geom_alluvium(aes(fill=Course)) +
geom_stratum() + geom_text(stat='stratum', aes(label=after_stat(stratum)))
```
```

4. Parallel Sets Alternative:

```
```R
library(ggforce)
ggplot(df) + geom_parallel_sets(aes(x=stage, id=id, split=node, value=Students, fill=Course))
```
```

5. Recommendation:

An Alluvial diagram (or grouped Parallel Sets) colored by the originating 'Course' is better. It clearly shows the drop-off rate (Certification = No) while allowing the viewer to track exactly which initial course leads to which specific Job Role without visual confusion.